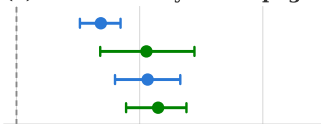


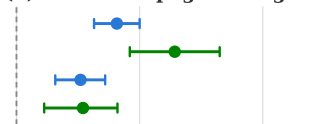
(a) Confirmatory below-page gate (post-fix)

vs commit-once · MMLB
vs commit-once · LDU
vs gold static · MMLB
vs gold static · LDU



(b) Historical page-level gate (pre-fix)

vs commit-once · MMLB
vs commit-once · LDU
vs gold static · MMLB
vs gold static · LDU



(c) Clean-cache re-run (24 docs)

vs commit-once (pooled)
vs gold static (pooled)



(d) 30-doc below-page replication

vs first-query commit · MMLB



0.0 0.2 0.4

Δ accuracy, fresh – comparator (document-clustered 95% CI)